

Full ablation comparison — how this analysis works

Conditions included: No memory, General facts, Theory of mind, Full KG

Exchange cadence in this run

Each agent now exchanges once per hour, not once per day: 100,800 rows over 30 days works out to 140 agents x 24 hours x 30 days. The original 'iteration' column in the raw file was a running exchange counter, not a day index. This report reindexes it to the day number (iteration = day_id) before computing entropy, left-leaning share, and effective clusters, so those three numbers describe a full-day population snapshot, not a single hourly exchange. Acceptance rate by stance distance and the transition/acceptance matrices are computed on every hourly exchange directly, since those are about individual conversations, not population state, and using every exchange gives more statistical power rather than less.

What this report measures

Each condition ran the same 140-agent population through the same simulation length, differing only in what memory the agents carry between conversations. Every exchange between two agents is logged as one row: who talked to whom, what each believed before, and what each believed after. This report reads those rows and computes four things, per condition:

1. Entropy: how spread out opinions are across the population on a given day. High means diverse, low means the population is converging on one view. Reported at the first and last day.
2. Left-leaning share: percent of the population holding a LEFT or FAR LEFT stance, at the start and end. This tracks directional drift specifically, separate from overall diversity.

Full ablation comparison — how this analysis works (continued)

3. Effective clusters: an estimate of how many distinct opinion groups remain at the end. Near 1 means everyone converged to roughly the same view. Near 5 means the population stayed spread across the full range.
4. Acceptance rate by stance distance: the probability an agent shifts toward its partner's stance, split out by how far apart the two agents started. This is the main test in this project: if memory reduces agreement bias, this rate should fall off faster as distance increases, compared to the no-memory condition.

How to read the tables and figures that follow

The summary table gives one row per condition with the four numbers above. The acceptance table breaks the fourth number out by exact stance distance, one column per condition, so the shapes of the curves can be compared directly. The figure pages after that show the same information as plots, which is usually the fastest way to spot a difference between conditions.

This report only computes and tabulates. It does not state whether the hypothesis is supported. Comparing these numbers against the hypothesis criteria is a separate, deliberate step, done by reading `paper/context/hypothesis_and_scope.md` alongside this report.

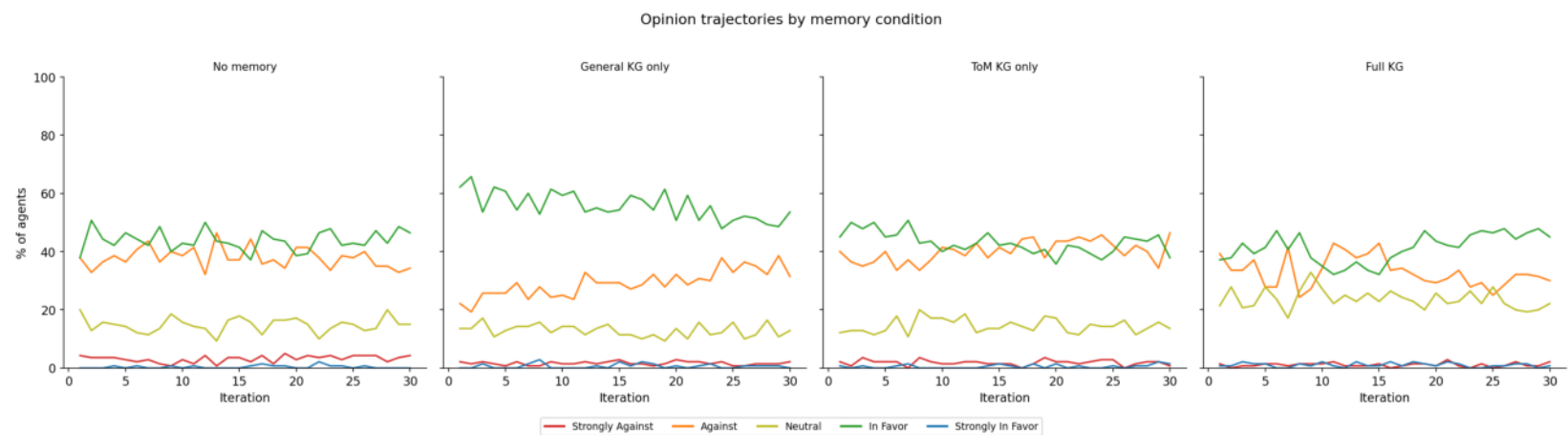
Summary

Condition	Days	Rows	Entropy start	Entropy end	Left start	Left end	Eff. clusters (end)
No memory	30	100800	1.7202	1.6487	42.1%	38.6%	2.88
General facts	30	100800	1.4180	1.5065	24.3%	33.6%	2.62
Theory of mind	30	100800	1.5863	1.5740	42.1%	47.1%	2.79
Full KG	30	100800	1.6750	1.6908	40.7%	32.1%	2.96

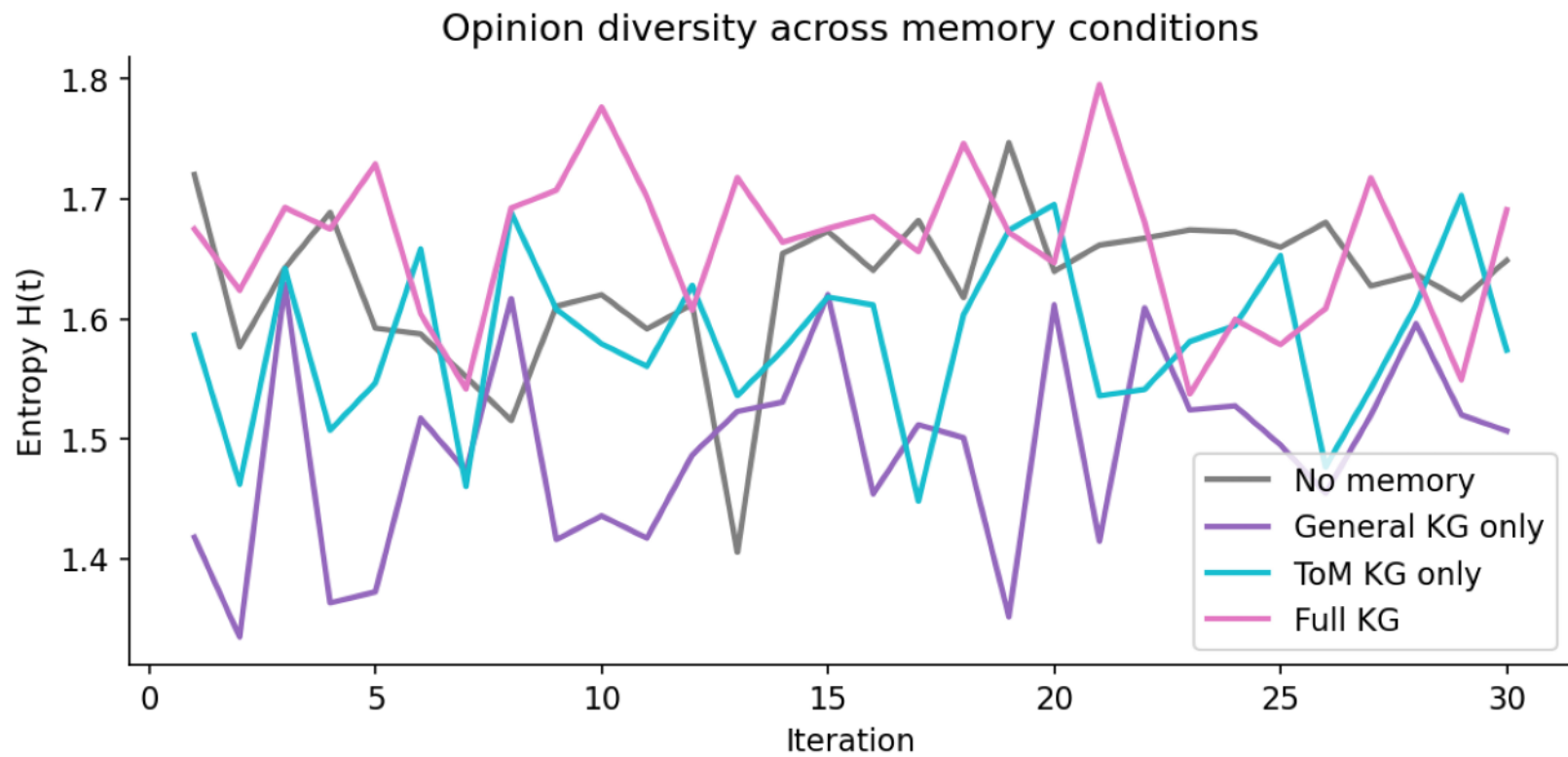
Acceptance rate by stance distance

Distance	No memory	General facts	Theory of mind	Full KG
-4	95.6%	95.0%	100.0%	95.8%
-3	22.2%	25.0%	26.3%	45.7%
-2	11.2%	10.3%	13.2%	23.6%
-1	23.2%	18.4%	24.9%	31.9%
1	27.4%	36.0%	30.1%	38.6%
2	9.1%	9.6%	9.8%	29.9%
3	55.2%	59.6%	55.7%	57.8%
4	66.7%	80.0%	64.7%	79.2%

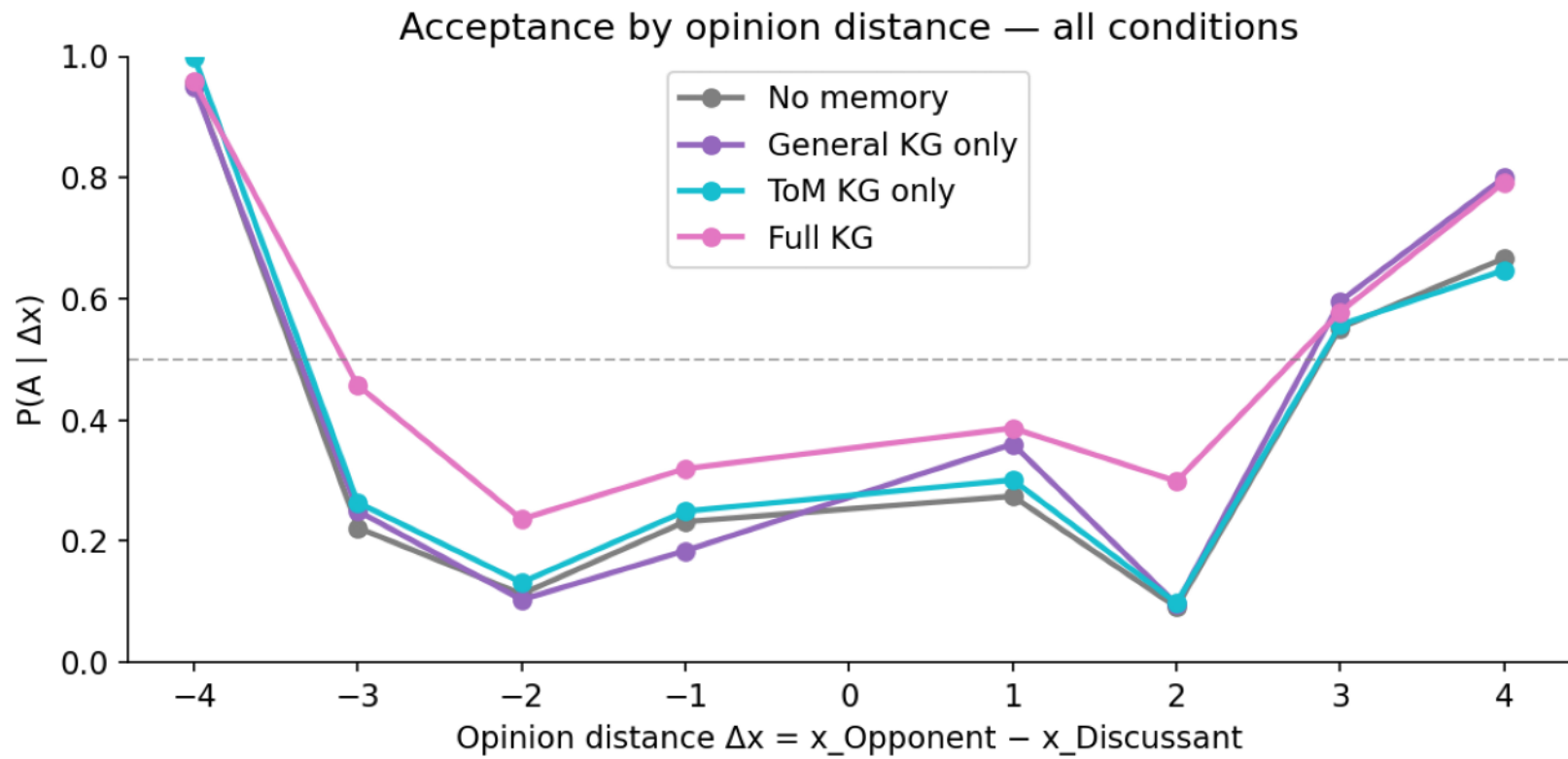
compare_trajectories.png



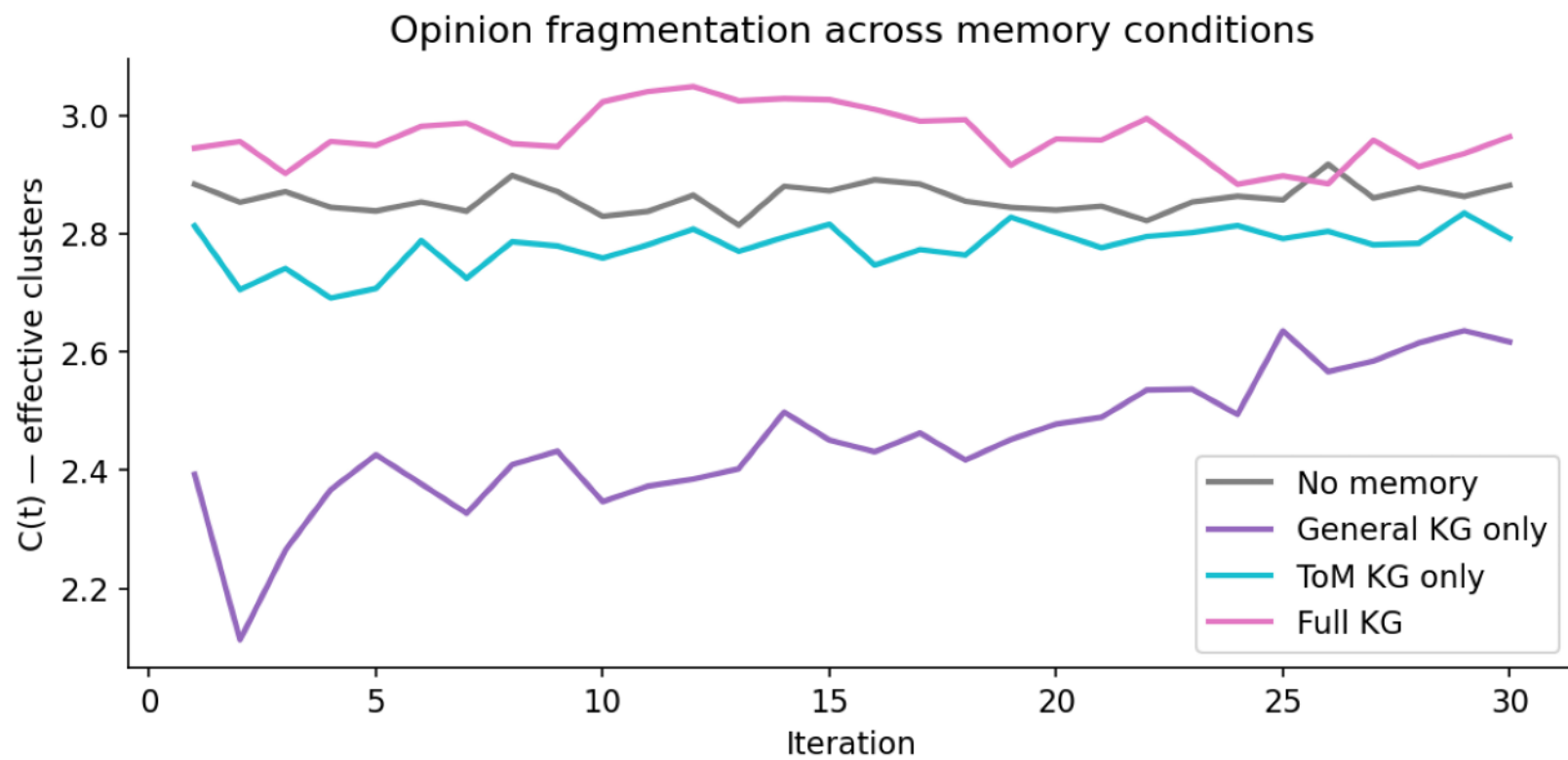
Raw stance distribution over time, one panel per condition. Shows directional drift directly: watch whether one color grows steadily at the expense of the others.



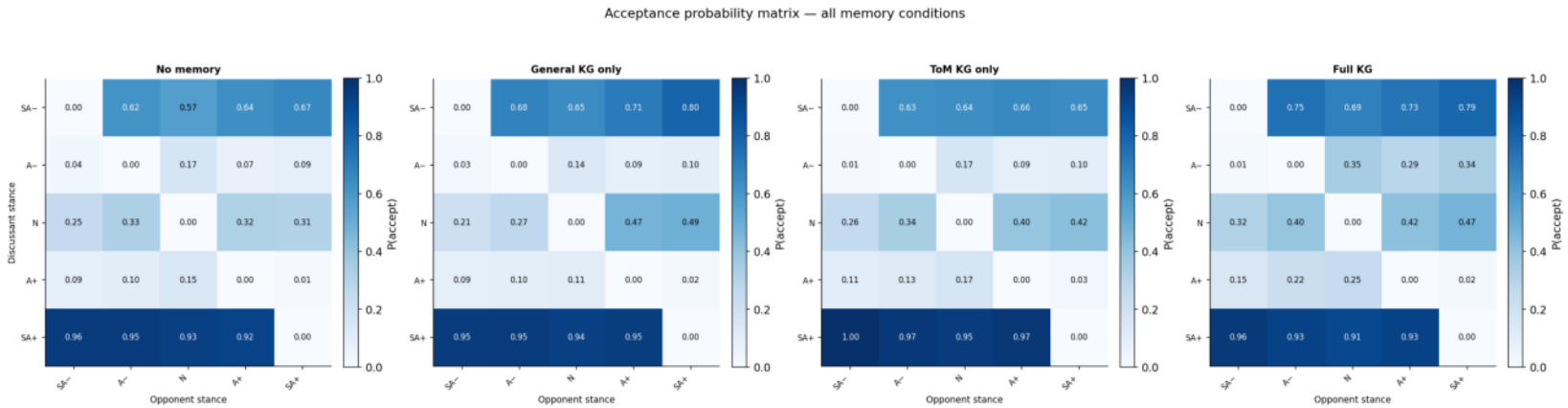
Population diversity over time, one line per condition. A condition that stays higher keeps opinions more spread out; a condition that drops fastest is converging toward consensus fastest.



The central test. Each line shows how likely an agent is to shift toward its partner's stance, as a function of how far apart the two started. A flatter line means that condition resists agreement with distant partners more than the baseline does.

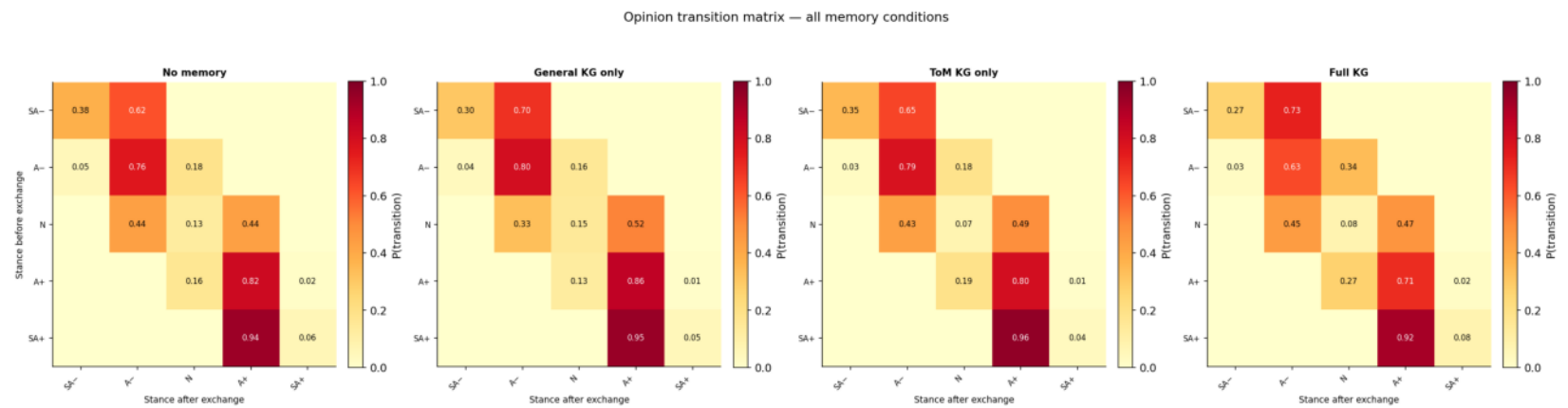


Estimated number of distinct opinion groups still present at the end of the run, per condition. Near 1 means the population converged to one view; near 5 means it stayed spread across the full scale.



Full pairwise detail behind the acceptance rate: probability of accepting a shift, broken down by the agent's own stance and the partner's stance, not just the distance between them.

compare_transition_matrix.png



Full pairwise detail on where agents actually ended up: probability of moving from each starting stance to each ending stance, per condition.